

CUSUM RAP Tool:

Administration and User Guide

1. App Overview

The CUSUM RAP (Reproducible Analytical Pipeline) Tool is a specialized Shiny application representing WS5.WP1.2 Product 1. It implements a CUSUM-based early detection algorithm on weekly case counts per spatial unit.

This tool is the final analytical step in the ecosystem and is designed to run sequentially:

1. **Data Collection:** Individual-level suspected case data is collected via **RapidCaseReport**.
2. **Aggregation:** Data is filtered and aggregated by week and spatial unit using **API-POP**.
3. **Analysis:** The aggregated CSV output from API-POP is used as the direct input for this CUSUM RAP Tool.

2. Administrator Setup & Credentials

Because this application relies on aggregated CSV uploads rather than live database connections, it does not require complex environment credentials. There are no `.Renvi` or `config.yaml` files to configure. The app can be run immediately upon deployment or packaged directly into a standalone executable.

3. Customization & Localization

Modifying the UI Design (config.css) The application's styling is controlled by the `config.css` file. Unlike other apps in this suite, the colors here are hardcoded directly into the elements rather than using `:root` variables. To change the app's branding, administrators must find and replace specific hex codes throughout the file:

- **Backgrounds & Navigation:** Look for `#4a4a4a` (dark gray) and `#333` (almost black) to change the main navigation and active tabs.
- **Accents & Text:** Look for `#616161` (standard text) and `#7a7a7a` (input backgrounds) to adjust the contrast.

Adding Countries & Regions (config.json) The spatial hierarchy for the analytical output is defined in a simplified `config.json` file. To add a new geographical configuration, simply add a new array defining the administrative levels from highest to lowest.

- Example for a new country: `"Brazil": { "levels": ["country", "state", "municipality"] }`

4. End-User Guide

Welcome to the CUSUM RAP Tool. This application allows you to run early-detection algorithms on your aggregated epidemiological data.

Step 1: Prepare Your Data Before using this tool, ensure you have exported your weekly aggregated suspected case counts (stratified by spatial unit) from the API-POP application. Save this file as a CSV.

Step 2: Upload and Configure Load your aggregated CSV file into the application. Use the sidebar menu to verify that the spatial variables (e.g., Province, Department) and time variables (e.g., Epi-week) are correctly mapped to your uploaded data.

Step 3: Run the Analysis Click the primary action button to execute the CUSUM algorithm. The pipeline will process the weekly counts and identify any spatial units where the reported suspected cases exceed the expected threshold, flagging potential early-stage outbreaks.

Step 4: Review and Export Review the generated visual charts and data tables. You can download the final analytical report using the download options provided in the interface.